

MINISTRY OF EDUCATION, CULTURE AND RESEARCH OF THE REPUBLIC OF MOLDOVA
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Report

Laboratory Work No. 1

Operating Systems

Topic:The OS as a resource manager

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OS Lab 1 Report

whoami

```
alex@ubuntu:~/os-lab1$ whoami
alex
```

The whoami command shows the username of the user currently logged into the system. In this case, the username is alex.

uname -a

```
alex@ubuntu:~/os-lab1$ uname -a
Linux ubuntu 7.0.0-38-generic #38-Ubuntu SMP
PREEMPT_DYNAMIC Fri Sep 4 09:10:14 UTC 2026 x86_64
GNU/Linux
```

The uname -a command displays detailed information about the operating system and kernel. Here, the system is running Ubuntu with a 64-bit (x86_64) Linux kernel.

uptime

```
alex@ubuntu:~/os-lab1$ uptime
16:02:55 up 8 min, 1 user, load average: 0.22, 0.15,
0.08
```

The uptime command shows how long the system has been running, how many users are currently logged in, and the system load average.

Part 1: Files and directories

The user(me) owns the files created , and group is also named after me.

```
alex@ubuntu:~/os-lab1/demo$ ls -l
total 8
-rw-rw-r-- 1 alex alex 24 Oct  4 16:16 note.txt
-rw-rw-r-- 1 alex alex 24 Oct  4 16:16 renamed.txt
alex@ubuntu:~/os-lab1/demo$ rm renamed.txt
```

After chmod 600 the 10 characters mean : the first character means file type, character 2-4 mean the access mode for the user, characters 5-7 mean the access mode for the group, and characters 8-10 mean the access mode for everyone else.

```
alex@ubuntu:~/os-lab1/demo$ ls -l note.txt
```

```
-rw-r--r-- 1 alex alex 24 Oct  4 16:16 note.txt
```

/os-lab1/demo$ the first directory holds demo directroy which holds the demo directory with note.txt file.

Part 2: Process management

Commands execution:

```
alex@ubuntu:~$ ps aux | head
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT
START   TIME COMMAND
root           1  0.0  0.4  25268 16624 ?        Ss
15:54   0:01 /usr/lib/systemd/systemd --switched-root
--system --deserialize=51
root           2  0.0  0.0      0     0 ?        S
15:54   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S
15:54   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<
15:54   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<
15:54   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<
15:54   0:00 [kworker/R-kvfree_rcu_reclaim]
root           7  0.0  0.0      0     0 ?        I<
15:54   0:00 [kworker/R-slub_flushwq]
root           8  0.0  0.0      0     0 ?        I<
15:54   0:00 [kworker/R-netns]
root          10  0.0  0.0      0     0 ?        I<
15:54   0:00 [kworker/0:0H-kblockd]
alex@ubuntu:~$ ps aux | wc -l
214
alex@ubuntu:~$ top

top - 16:43:06 up 48 min,  1 user,  load average:
0.16, 0.31, 0.23
Tasks: 212 total,   1 running, 211 sleeping,   0
stopped,   0 zombie
%Cpu(s):  0.8 us,  0.8 sy,  0.0 ni, 98.4 id,  0.0 wa,
0.0 hi,  0.0 si
MiB Mem :   3302.8 total,    402.5 free,   1538.6
used,   1624.5 buff/
MiB Swap:      0.0 total,      0.0 free,      0.0
used.   1764.2 avail
```

```
    PID USER      PR  NI    VIRT    RES    SHR S  %CPU
%MEM
      4 root       0 -20       0      0      0 I   0.0
0.0
      5 root       0 -20       0      0      0 I   0.0
0.0
      6 root       0 -20       0      0      0 I   0.0
0.0
      7 root       0 -20       0      0      0 I   0.0
0.0
      8 root       0 -20       0      0      0 I   0.0
0.0
     10 root       0 -20       0      0      0 I   0.0
0.0
     12 root      20   0       0      0      0 I   0.0
0.0
     13 root       0 -20       0      0      0 I   0.0
0.0
     14 root      20   0       0      0      0 S   0.0
0.0
     15 root      20   0       0      0      0 I   0.0
0.0
     16 root      20   0       0      0      0 S   0.0
0.0
     17 root      20   0       0      0      0 S   0.0
0.0
     18 root      rt   0       0      0      0 S   0.0
0.0
     19 root      20   0       0      0      0 S   0.0
0.0
alex@ubuntu:~$ sleep 300 &
[1] 5790
alex@ubuntu:~$ jobs
[1]+  Running                 sleep 300 &
alex@ubuntu:~$ ps -ef | grep sleep
alex        5790    4745  0 16:43 pts/0    00:00:00
sleep 300
alex        5794    4745  0 16:44 pts/0    00:00:00
grep --color=auto sleep
alex@ubuntu:~$ kill <PID>
bash: syntax error near unexpected token `newline'
alex@ubuntu:~$ ^C
alex@ubuntu:~$ kill <PID>
bash: syntax error near unexpected token `newline'
```

```
alex@ubuntu:~$ kill <5790>
bash: syntax error near unexpected token `5790'
alex@ubuntu:~$ kill 5790
[1]+  Terminated                 sleep 300
alex@ubuntu:~$ sleep 300 &
[1] 5810
alex@ubuntu:~$ PID=$!
alex@ubuntu:~$ ls /proc/$PID/
arch_status         fdinfo              net
setgroups
attr                gid_map             ns
smaps
autogroup           io                  numa_maps
smaps_rollup
auxv                ksm_merging_pages   oom_adj
stack
cgroup              ksm_stat            oom_score
stat
clear_refs          latency             oom_score_adj
statm
cmdline             limits              pagemap
status
comm                loginuid            patch_state
syscall
coredump_filter     map_files           personality
task
cpu_resctrl_groups  maps                projid_map
timens_offsets
cwd                 mem                 root
timers
environ             mountinfo           sched
timerslack_ns
exe                 mounts              schedstat
uid_map
fd                  mountstats          sessionid
wchan
alex@ubuntu:~$ cat /proc/$PID/status | head -20
Name:    sleep
Umask:    0002
State:    S (sleeping)
Tgid:    5810
Ngid:    0
Pid:    5810
PPid:    4745
TracerPid:    0
```

```
Uid:    1000    1000    1000    1000
Gid:    1000    1000    1000    1000
FDSize:    256
Groups:    4 24 27 30 46 100 111 114 1000
NStgid:    5810
NSpid:    5810
NSpgid:    5810
NSsid:    4745
Kthread:    0
VmPeak:       16112 kB
VmSize:       16112 kB
VmLck:           0 kB
alex@ubuntu:~$ kill $PID
[1]+  Terminated                 sleep 300
```

Observe

the first process is sleep 300 and its PID is 5790

```
[1]+  Running                    sleep 300 &
alex@ubuntu:~$ ps -ef | grep sleep
alex        5790    4745  0 16:43 pts/0    00:00:00
sleep 300
alex        5794    4745  0 16:44 pts/0    00:00:00
grep --color=auto sleep
```

213 processes

```
alex@ubuntu:~$ ps aux | wc -l
213
```

the state says sleeping, which means the process is alive but its waiting.

```
State:    S (sleeping)
```

Part 3: Memory

Commands Execution:

```
alex@ubuntu:~$ free -h
               total        used        free
shared  buff/cache   available
Mem:           3.2Gi       1.5Gi       378Mi
36Mi       1.6Gi       1.7Gi
Swap:             0B          0B          0B
alex@ubuntu:~$ cat /proc/meminfo | head -6
MemTotal:        3382032 kB
```

```
MemFree:          312768 kB
MemAvailable:    1734320 kB
Buffers:           54100 kB
Cached:          1575860 kB
SwapCached:            0 kB
alex@ubuntu:~$ sleep 300 & PID=$!
[1] 6087
alex@ubuntu:~$ grep VmRSS /proc/$PID/status
VmRSS:        7508 kB
alex@ubuntu:~$ kill $PID
[1]+  Terminated              sleep 300
alex@ubuntu:~$
```

Observe

the VM has 3.2 Gi of Ram and 378 Mi is free

```
total        used        free      shared  buff/cache
available
Mem:           3.2Gi       1.5Gi       378Mi
36Mi       1.6Gi       1.7Gi
Swap:             0B          0B          0B
```

Swap: disk space the OS uses as overflow when RAM fills up. Inactive memory pages get moved there. It's slower than RAM. It shows 0's probably because this machine uses none.

```
Swap:             0B          0B          0B
```

The sleep process uses 7508 kb of ram and that kinda of surprises me because the process doesnt do anything but wait

```
alex@ubuntu:~$ grep VmRSS /proc/$PID/status
VmRSS:        7508 kB
```

Part 4: Devices and storage

Command output:

```
alex@ubuntu:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           661M  2.1M  659M   1% /run
/dev/sda2       9.8G  6.8G  2.6G  74% /
tmpfs           1.7G     0  1.7G   0% /dev/shm
none            1.0M     0  1.0M   0%
/run/credentials/systemd-journald.service
```

```
tmpfs           1.7G  8.0K  1.7G   1% /tmp
none            1.0M     0  1.0M   0%
/run/credentials/systemd-resolved.service
tmpfs           331M   80K  331M   1% /run/user/1000
/dev/sr0         51M   51M     0 100%
/run/media/alex/VBox_GAs_7.2.20
alex@ubuntu:~$ lsblk
NAME   MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0    7:0    0     4K  1 loop /snap/bare/5
loop1    7:1    0  66.8M  1 loop /snap/core24/1643
loop2    7:2    0  14.2M  1 loop
/snap/desktop-security-center/206
loop3    7:3    0    20M  1 loop
/snap/desktop-security-center/151
loop4    7:4    0 260.3M  1 loop /snap/firefox/8763
loop5    7:5    0 614.5M  1 loop
/snap/gnome-46-2404/164
loop6    7:6    0  16.5M  1 loop
/snap/firmware-updater/226
loop7    7:7    0  91.7M  1 loop
/snap/gtk-common-themes/1535
loop8    7:8    0   1.5M  1 loop /snap/hwctl/131
loop9    7:9    0   1.5M  1 loop /snap/hwctl/123
loop10   7:10   0   402M  1 loop /snap/mesa-2404/1839
loop11   7:11   0  18.8M  1 loop
/snap/prompting-client/222
loop12   7:12   0  11.8M  1 loop /snap/snap-store/1390
loop13   7:13   0  18.4M  1 loop
/snap/prompting-client/228
loop14   7:14   0  50.1M  1 loop /snap/snapd/27710
loop15   7:15   0   828K  1 loop
/snap/snapd-desktop-integration/391
sda      8:0    0  10.1G  0 disk
├─sda1   8:1    0     1M  0 part
└─sda2   8:2    0    10G  0 part /
sr0     11:0    1  50.7M  1 rom
/run/media/alex/VBox_GAs_7.2.20
alex@ubuntu:~$ du -sh ~/os-lab1
12K    /home/alex/os-lab1
alex@ubuntu:~$ ls -l /dev | head
total 0
crw-r--r--  1 root    root     10, 235 Oct  4 15:54
autofs
drwxr-xr-x  2 root    root         460 Oct  4 16:48
block
```

```
drwxr-xr-x  2 root    root          80 Oct  4 15:54 bsg
crw-------  1 root    root     10, 234 Oct  4 15:54
btrfs-control
drwxr-xr-x  3 root    root          60 Oct  4 15:54 bus
lrwxrwxrwx+ 1 root    root           3 Oct  4 15:54
cdrom -> sr0
drwxr-xr-x  2 root    root        3740 Oct  4 17:14
char
crw-------  1 root    tty        5,   1 Oct  4 15:54
console
lrwxrwxrwx  1 root    root          11 Oct  4 15:54
core -> /proc/kcore
alex@ubuntu:~$ mount | head
tmpfs on /run type tmpfs
(rw,nosuid,nodev,size=676408k,nr_inodes=819200,mode=75
5,inode64)
/dev/sda2 on / type ext4 (rw,relatime)
devtmpfs on /dev type devtmpfs
(rw,nosuid,size=731316k,nr_inodes=182829,mode=755,inod
e64)
tmpfs on /dev/shm type tmpfs
(rw,nosuid,nodev,inode64,usrquota)
devpts on /dev/pts type devpts
(rw,nosuid,noexec,relatime,gid=5,mode=600,ptmxmode=000
)
sysfs on /sys type sysfs
(rw,nosuid,nodev,noexec,relatime)
securityfs on /sys/kernel/security type securityfs
(rw,nosuid,nodev,noexec,relatime)
cgroup2 on /sys/fs/cgroup type cgroup2
(rw,nosuid,nodev,noexec,relatime,nsdelegate,memory_rec
ursiveprot,memory_hugetlb_accounting)
none on /sys/fs/pstore type pstore
(rw,nosuid,nodev,noexec,relatime)
bpf on /sys/fs/bpf type bpf
(rw,nosuid,nodev,noexec,relatime,mode=700)
```

Observe

My root files system is mounted on /dev/sda2

A /dev entry and what it stands for, for example: /dev/sda is the first physical (or virtual) hard disk /dev/null discards anything written to it /dev/tty is terminal

"Everything is a file": write it in your own words, something like: disks, terminals and even process info appear as entries in the filesystem, so the same operations (open, read, write) work on all of them, rather than needing a different interface for each.

Conclusion

During this laboratory work, I gained a basic understanding of how the xv6 operating system works. I built and launched xv6 using QEMU, used its shell to run basic commands, and explored how programs such as `cat`, `echo`, and `grep` work. I also examined the `user/` and `kernel/` directories and learned how user programs communicate with the kernel through system calls such as `read()` and `write()`. Overall, the lab helped me understand the separation between user programs and the operating system kernel and how the kernel provides services to those programs.